

Set Theory and Relations

1. Sets and Their Operations

Definition: A set is a collection of distinct elements.

Description of a set

Formal: We use precise mathematical notation. - *Examples:*

$$B = \{x \mid x \in \mathbb{N}, x \bmod 2 = 0\}$$

$$S = \{x \mid x = n^2, n \in \mathbb{N}\}$$

Power Set: The set of all subsets of a given set.

Example in Python:

```
def power_set(s):
    x = len(s)
    subsets = []
    for i in range(1 << x):
        subsets.append([s[j] for j in range(x) if (i & (1 << j))])
    return subsets

print(power_set({1, 2, 3}))
```

Operations:

Union: $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$

Intersection: $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

Cartesian Product: $A \times B = \{(a, b) \mid a \in A, b \in B\}$